

What's the Android Open Accessory Protocol All About?



This article helps you to start using the Android Open Accessory (AOA) protocol, and also illustrates how to make use of a simple Linux machine (PC, notebook or any embedded board) to communicate with Android apps via USB.

The Android Open Accessory (AOA) protocol allows simple USB communication with two bulk end points between two nodes, which are:

- An external hardware device with USB host capabilities and peripheral support known as an Accessory Development Kit (ADK) or Accessory Host, from now onwards.
- An Android gadget, which works in device mode.

The ADK host should be capable of supplying 500mA at 5V power to the device. We know that most Android gadgets come with USB device capabilities in order to support data transfer and debugging (adb); so this protocol chooses external hardware in the host mode and an Android machine in device mode. This eliminates the need for host support on the Android side as very few gadgets come with such capabilities. Also, there is no need to root the device for any changes to the Android kernel, as peripheral support is taken care of by external hardware.

History

AOA version 1.0 was launched at Google I/O 2011 with

support from Android versions 3.1 onwards, and with an add-on library for the Android versions 2.3.4 and above. This was demonstrated with a custom shield on top of an Atmega-based Arduino Mega board.

Later in 2012, version 2.0 of this protocol was released at Google I/O 2012 with additional support for communication over Bluetooth, apart from USB. Also, two new features were added with Audio and Human Interface Devices (HID) capabilities from the ADK side. This was demonstrated with a custom kit made from the ARM Cortex-m3-based Arduino Due board with open hardware specs, popularly known as the ADK 2012 kit.

Supporting hardware

Here is a list of some boards which can act like ADK, with the availability of sample code or libraries:

- Arduino Mega board
- Arduino Due board
- AOA kit from Embedded Artists
- Microchip PIC24F series MCUs
- Sparkfun IOIO
- FTDI Vinculum - II